INTERNETWORKING

CIS*3210

Fall 2026

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Based on notes for Computer Networking: A Top-Down Approach, 9th ed., by J. Kurose and K. Ross

Learning Outcomes

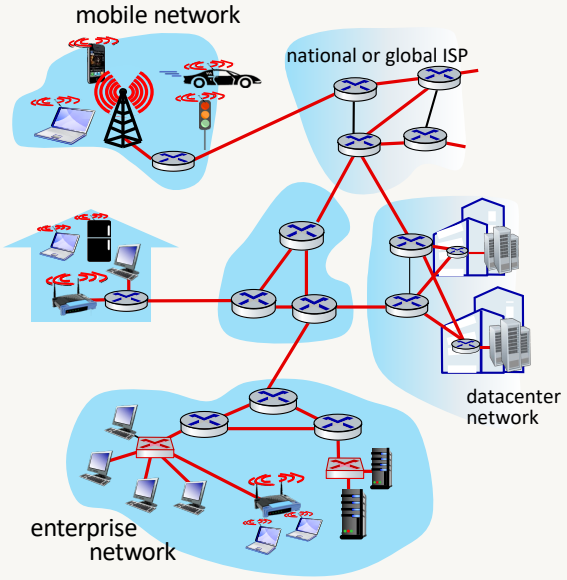


- Explain why different network technologies are needed
- Describe how routers and internetworking provide universal service across heterogeneous networks
- Distinguish between an intranet and the global Internet

Motivation for Internetworking



- Each network technology is built to meet specific constraints
  - LAN (Local Area Network): optimized for high-speed communication over short distances
  - WAN (Wide Area Network): optimized for communication over long distances
- No single networking technology meets every requirement
  - Large organizations with varied networking needs often require multiple physical networks
  - Selecting the best network type for each task results in multiple network technologies in use
  - Example:
    - Ethernet (LAN) for connecting computers within the same site
    - Leased data circuits (WAN) for linking sites in different cities



Motivation for Internetworking



Discussion Question:

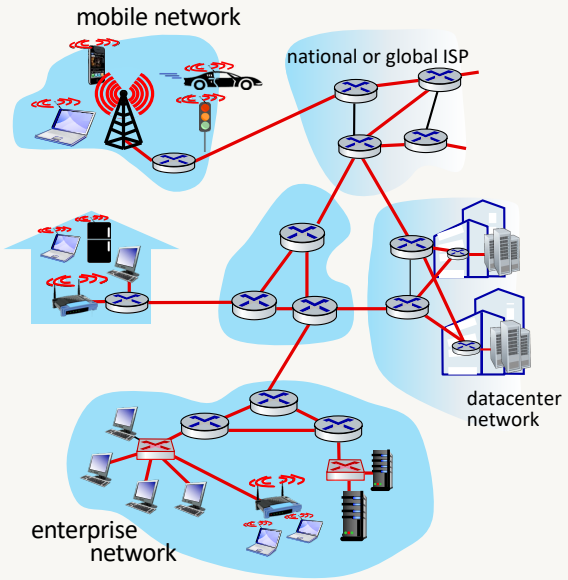
A company uses Ethernet inside each office and leased circuits between cities. Why would replacing both with one technology potentially make the system worse, even if that technology can technically span both distances?

A technology that spans both distances may not be optimized for either environment. The company could lose the low cost and high local speed of Ethernet or the reach and service guarantees of leased circuits. Universal service requires interoperability, not identical physical networks.

The Concept of Universal Service



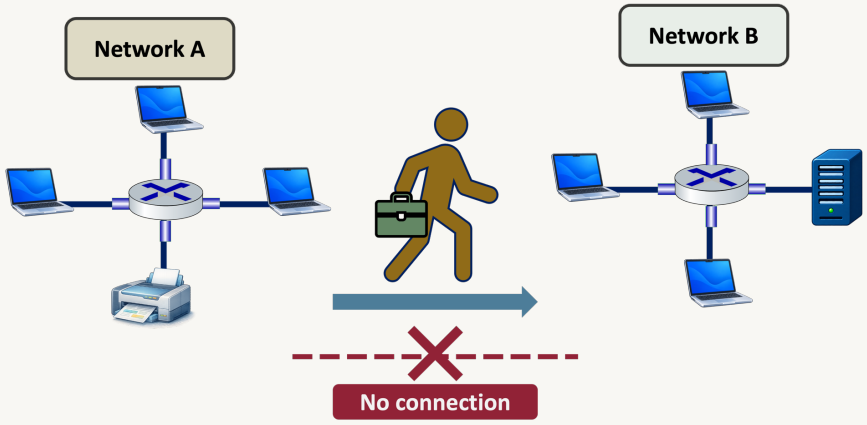
- Most modern computer communication systems allow communication between any two computers
  - Analogous to the way a telephone system provides communication between any two telephones, known as **universal service**
- With universal service
  - A user on any computer in any organization can send messages or data to any other user



The Concept of Universal Service



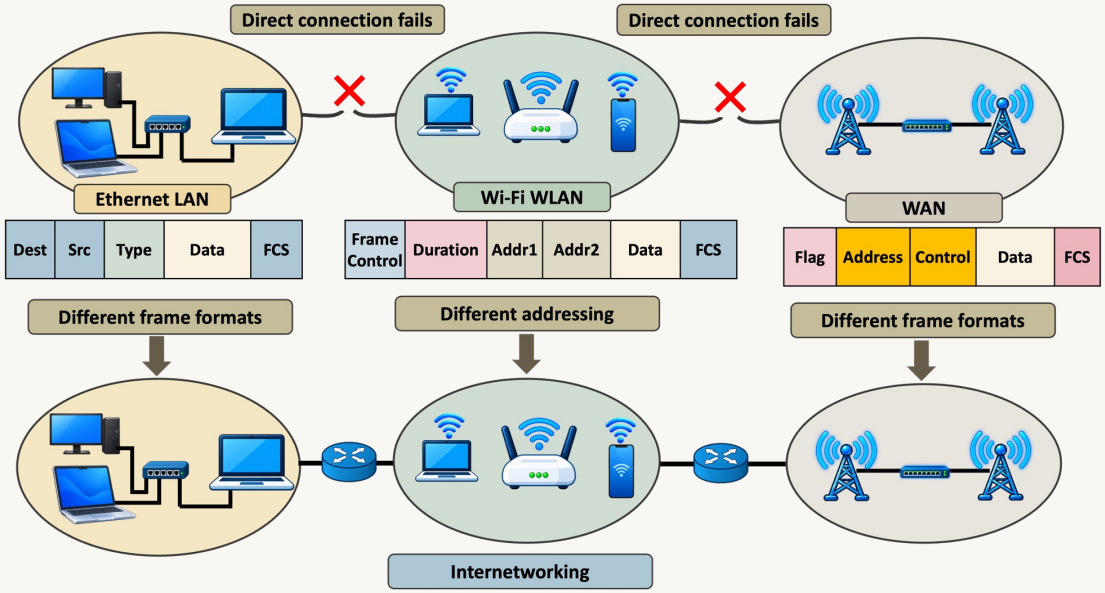
- The main issue with having multiple networks is that a computer can only communicate with others on the same network
  - This problem became clear in the 1970s when large organizations started using multiple networks
  - Each network acted as a separate “island” within the organization
- In many early setups:
  - Each computer was connected to just one network
  - Employees had to select the right computer for each task
  - Some had access to multiple screens and keyboards
  - Sending messages across networks required physically moving to a different computer
  - This setup reduced both user satisfaction and productivity



Universal Service in a Heterogeneous World



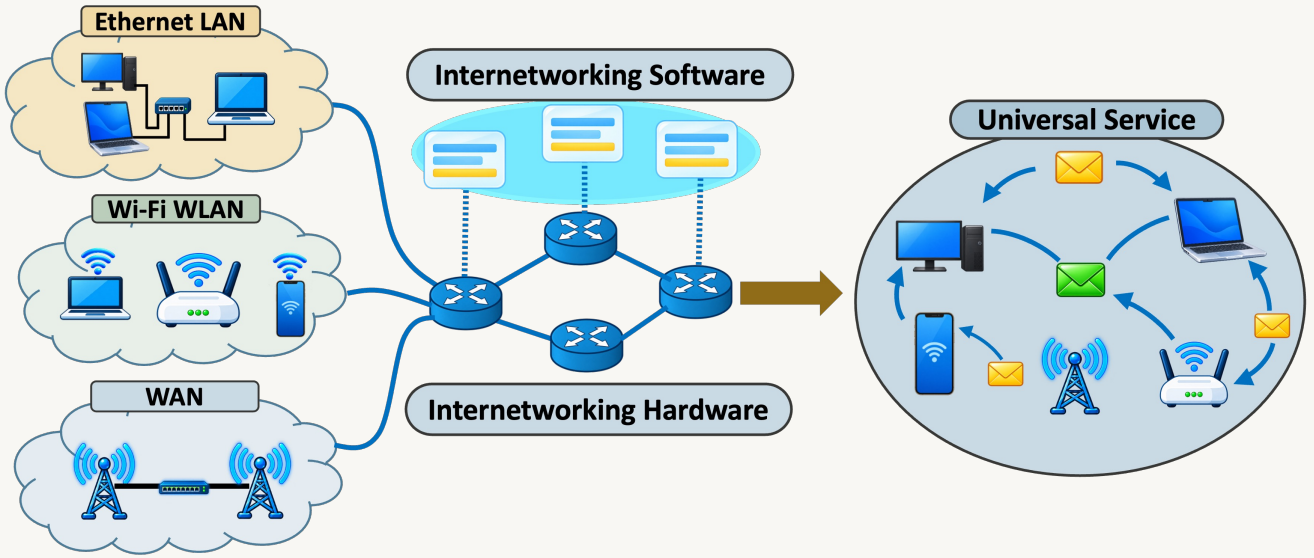
- The question arises: Does universal service require everyone to use a single network technology?
  - Is universal service possible across networks that use different technologies?
- Incompatibilities prevent building a large network simply by connecting network wires together
- Bridging and similar extension techniques cannot be applied to heterogeneous network technologies
  - Each technology has its own packet format and addressing method
  - A frame created for one network technology cannot be transmitted on a network using a different technology



Internetworking



- **Internetworking** enables universal service across different network technologies
  - Hardware interconnects the networks
  - Software enables communication across them
  - The connected networks form an internetwork, or **internet**
- An internet can vary in size
  - It may contain a few networks or, like the global **Internet**, tens of thousands
  - Each network may contain anywhere from no computers to hundreds



Internetworking



Discussion Question:

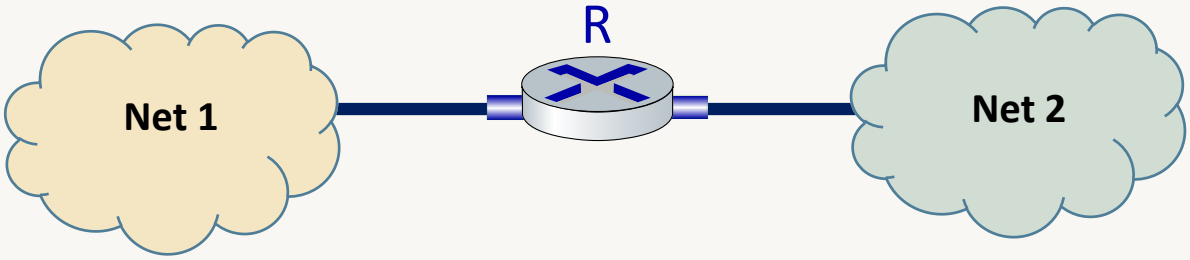
A designer says, 'If two networks cannot exchange native frames, they cannot be part of one internet.' Evaluate the statement.

The statement confuses link-level compatibility with internetwork communication. Networks can remain physically different while routers carry a common internetwork packet by re-encapsulating it in the native frame required on each link.

Physical Network Connection with Routers



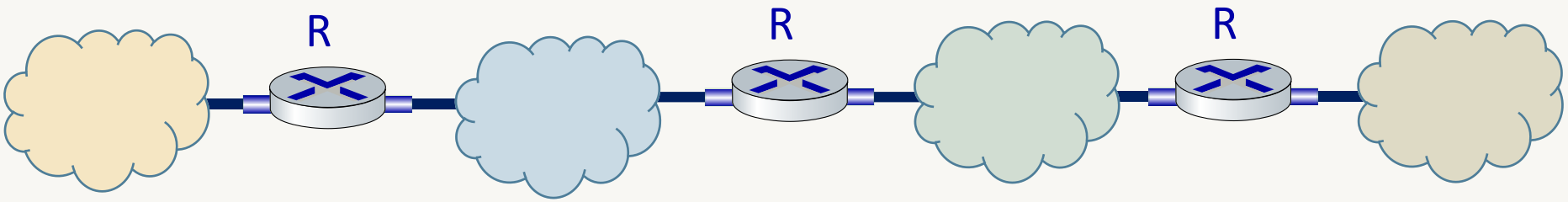
- A **router** is the primary device used to connect heterogeneous networks
  - It is a dedicated hardware system with a processor, memory, and an interface for each network
- A router can connect networks using different technologies
  - Two LANs, a LAN and a WAN, or two WANs
  - For example, an Ethernet network and a Wi-Fi network



Internet Architecture



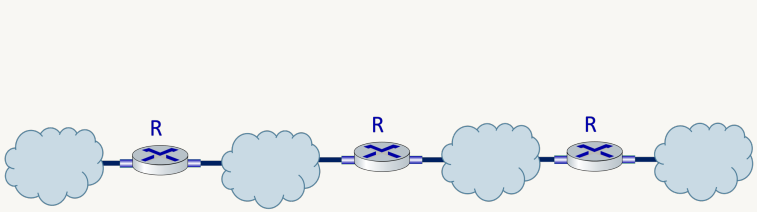
- Organizations choose appropriate network technologies for each need and use routers to connect them
- Three routers can be used to connect four arbitrary physical networks into an internet

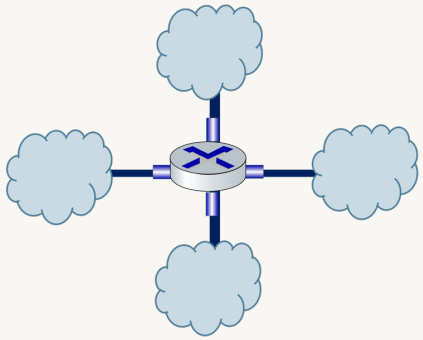


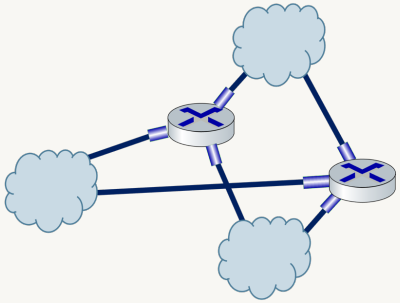
Internet Architecture



- Commercial routers can connect more than two networks
  - A single router could connect all four networks
- Organizations usually deploy multiple routers for
  - Load balancing and improved performance
  - Redundancy and protection against a single point of failure
- Protocol software monitors connections and reroutes traffic when a network or router fails







Internet Architecture



- An organization must select a design that satisfies its needs for reliability, capacity, and cost
- The specific internet topology chosen often depends on
  - Bandwidth of the physical networks
  - Expected traffic
  - Organization's reliability requirements
  - Cost
  - Performance of available router hardware

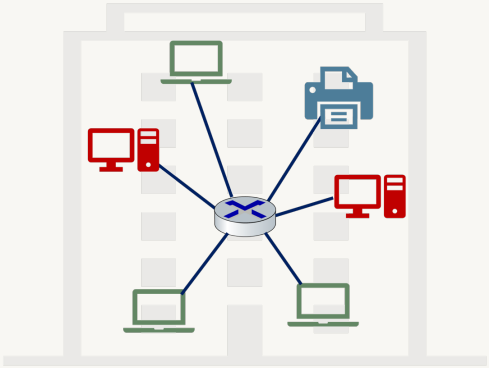
Intranet and Internet



- Intranet
  - Owned by a private organization
  - Intended for use only by the organization's employees
- Internet
  - Refers to the global Internet and its associated protocols
- Distinction between intranet and Internet is blurred due to
  - Organizations use the same equipment and protocol software for intranets as ISPs do for the global Internet
  - Organizational intranets often connect to the global Internet through gateways



Internet



Intranet

Achieving Universal Service



- Routers must agree to forward information
- The task is complex because
  - Frame formats and addressing schemes can vary between underlying networks
- Protocol software enables universal service
- Internet protocols resolve differences in frame formats and physical addresses
  - This allows communication between networks using different technologies

(When written with an uppercase **I**, the term Internet refers to the current global Internet and the associated protocols)

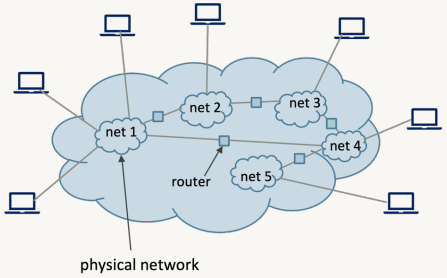
A Virtual Network



- The Internet gives the impression of a single seamless communication system
  - A mix of hardware and software creates the illusion of a uniform network
- Internet software conceals details such as
  - Physical network connections
  - Physical addresses
  - Routing information
- Users and application programs are not expected to know about the underlying physical networks or connecting routers
- An internet is considered a virtual network system
  - Because it represents an abstraction of the communication system

The illusion of a single network provided to users and applications





The underlying physical structure with routers interconnecting networks

The Internet: Nuts and Bolts





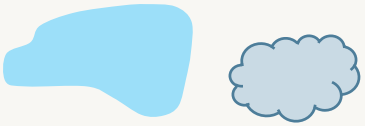
- Billions of Internet-connected devices
  - Also called **end systems** or hosts
  - Located at the Internet's **edge**, running applications



- Packet switches are devices that transfer data in chunks called packets
  - Common types include routers and network switches
  - These devices are responsible for directing packets through the network toward their destination
  - A switch connects devices within a LAN, while a router connects multiple networks together



- Communication links:
  - optical fibre, copper cables (e.g., Ethernet), wireless radio signals (e.g., Wi-Fi, cellular), satellite connections
  - Transmission rate: Measured in bandwidth (bits per second)



- Networks consist of:
  - A group of interconnected devices
  - Communication links (e.g., fibre, wireless)
  - Routers and other networking hardware

Internet-Connected Devices





Amazon Echo



Internet refrigerator



IP picture frame



Pacemaker & Monitor



Tweet-a-watt: monitor energy use

bikes





cars



Web-enabled toaster + weather forecaster



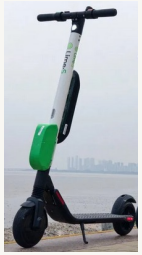
Slingbox: remote control cable TV



Security Camera



AR devices



scooters



Fitbit



Internet phones



Gaming devices



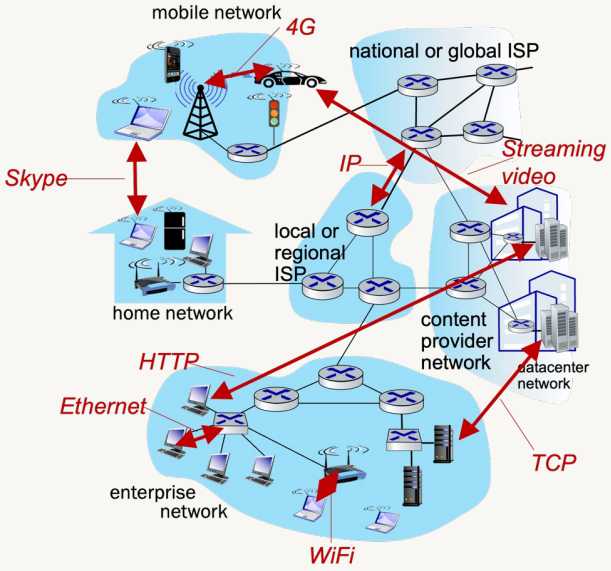
sensorized bed mattress

Others?

The Internet: Nuts and Bolts



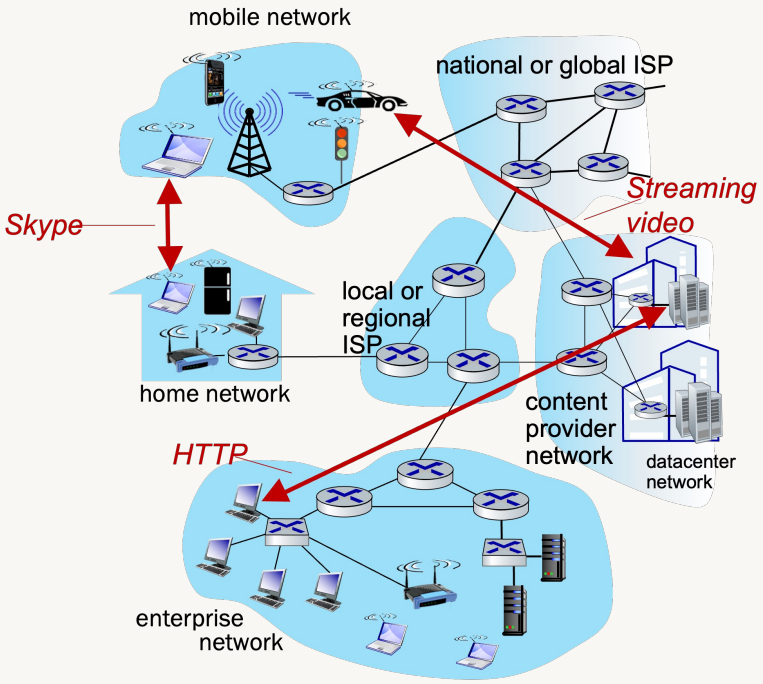
- The **Internet** is a global system of interconnected networks
  - Connects independent networks and devices worldwide
  - Built through interconnected Internet Service Providers (ISPs)
- **Protocols and protocol suites** define how devices exchange data
  - Examples include HTTP and TCP/IP
- **Networking standards and technologies** provide wired and wireless connectivity
  - Examples include Ethernet, Wi-Fi, and 4G/5G
  - Support applications such as video streaming and Zoom
- **Internet standards** ensure interoperability
  - **RFCs:** Documents that define protocols and technical standards
  - **IETF:** Organization that develops and maintains many Internet standards



The Internet: A Services View



- Internet infrastructure supports a wide range of applications:
  - web browsing, streaming video, multimedia teleconferencing, email, online games, e-commerce, social media, interconnected appliances
- Offers a programming interface for distributed applications:
  - Provides “hooks” that enable sending and receiving apps to connect and use Internet transport services
  - Supplies service options similar to how postal services offer different delivery choices

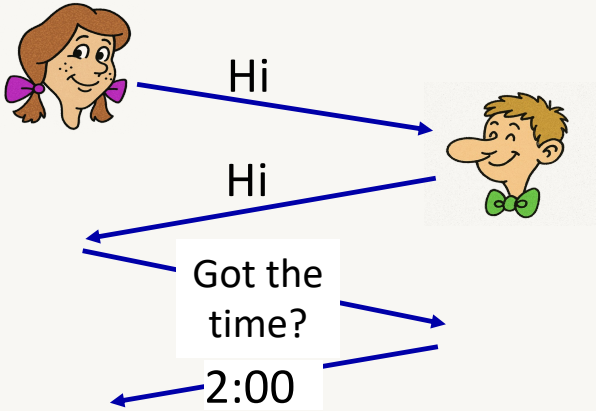


What is a Protocol?

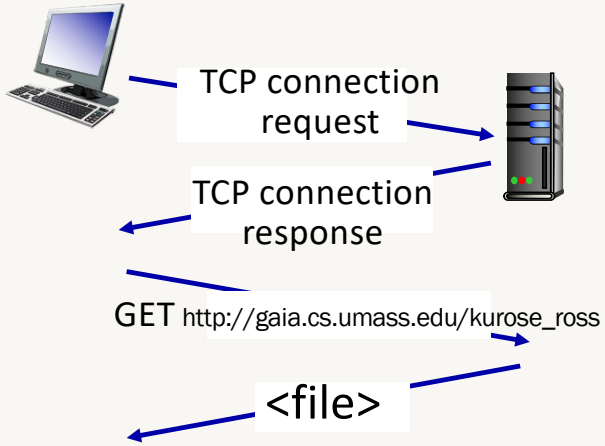


A human protocol and a computer network protocol:

Initiating a respectful communication



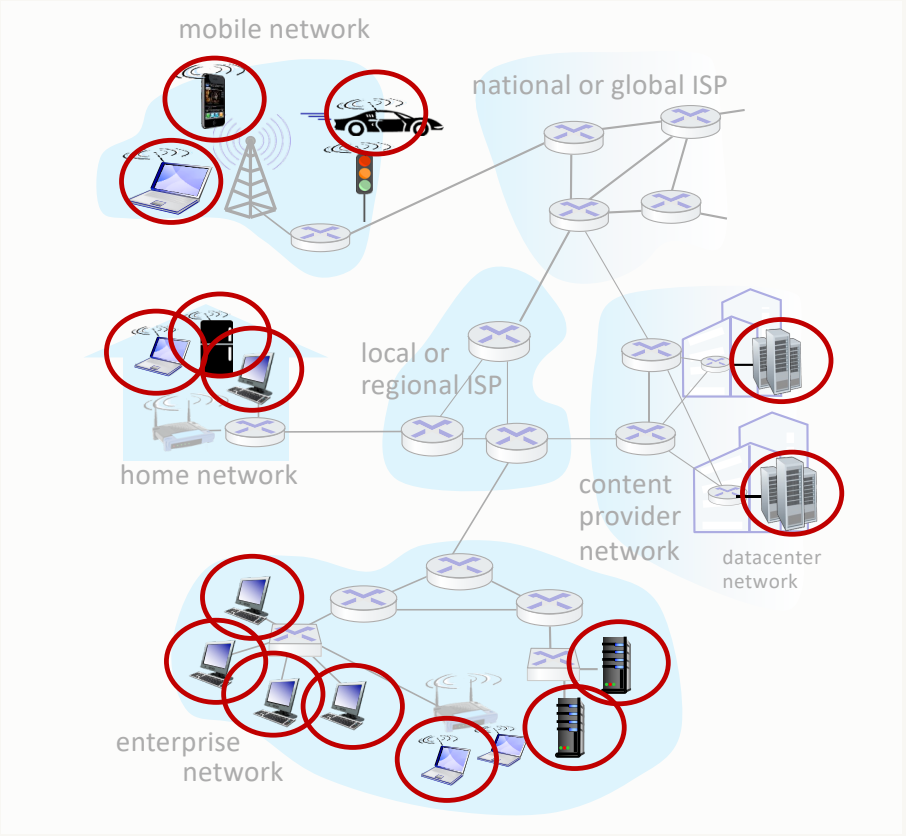
Exchanging messages and taking actions



Internet Structure



- **Network edge**
  - Hosts include clients and servers
  - Referred to as hosts because they host application programs such as a Web browser program, a Web server program, an e-mail client program, or an e-mail server program
  - Servers are often located in data centers

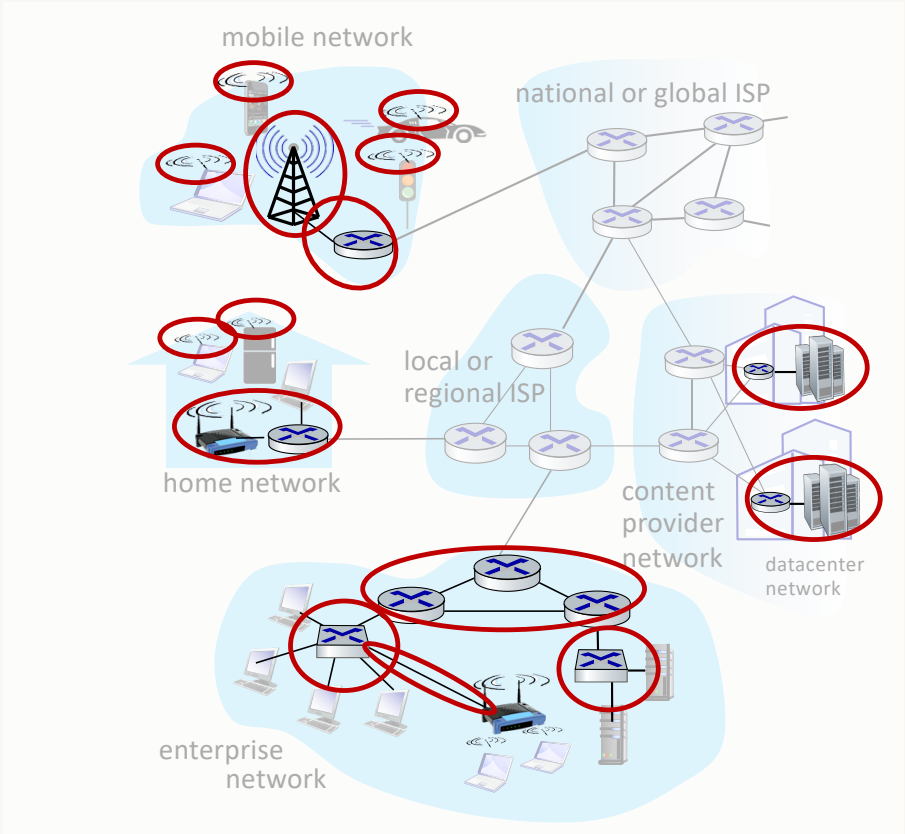


Internet Structure



Access Networks and Physical Media:

- Access network: the network that directly links an **end system** to the first router (**edge router**) on the path to another distant **end** system
- Use wired or wireless communication links

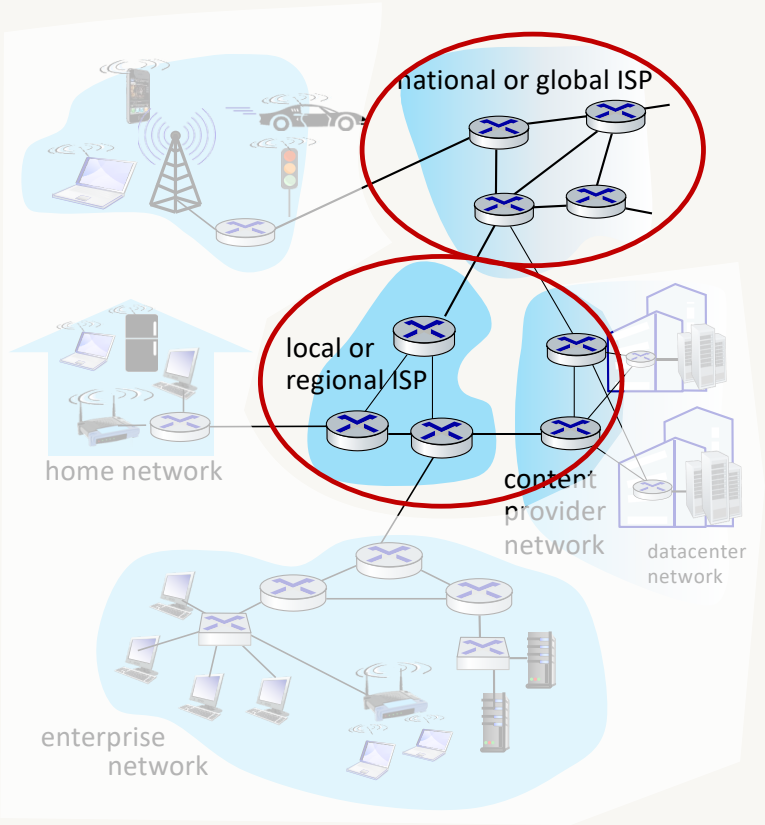


Internet Structure



Network core:

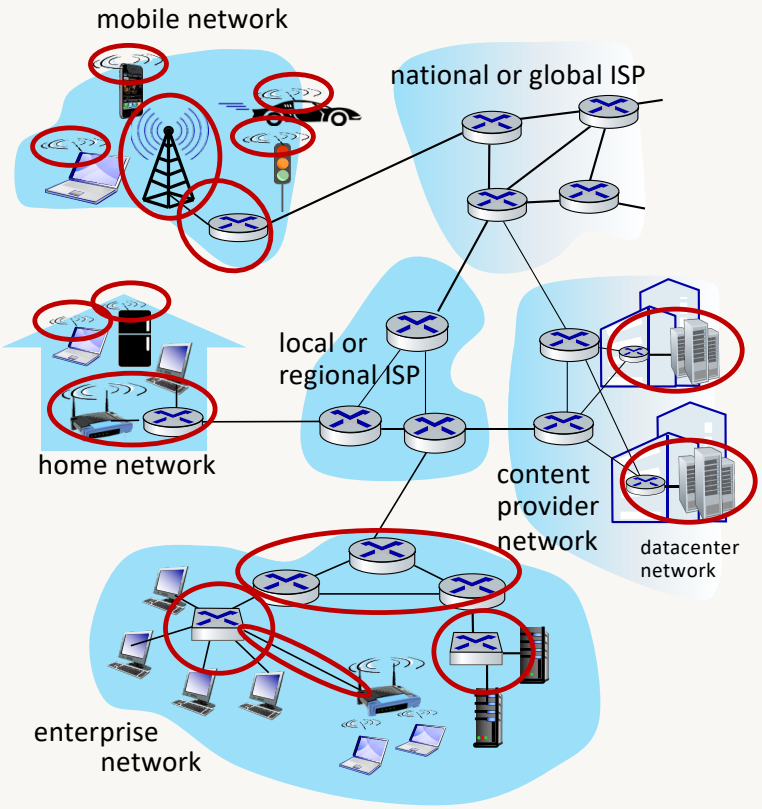
- Network core: the interconnected mesh of packet switches and links that connects the Internet's **end** systems
- interconnected routers
- network of networks



Physical Media



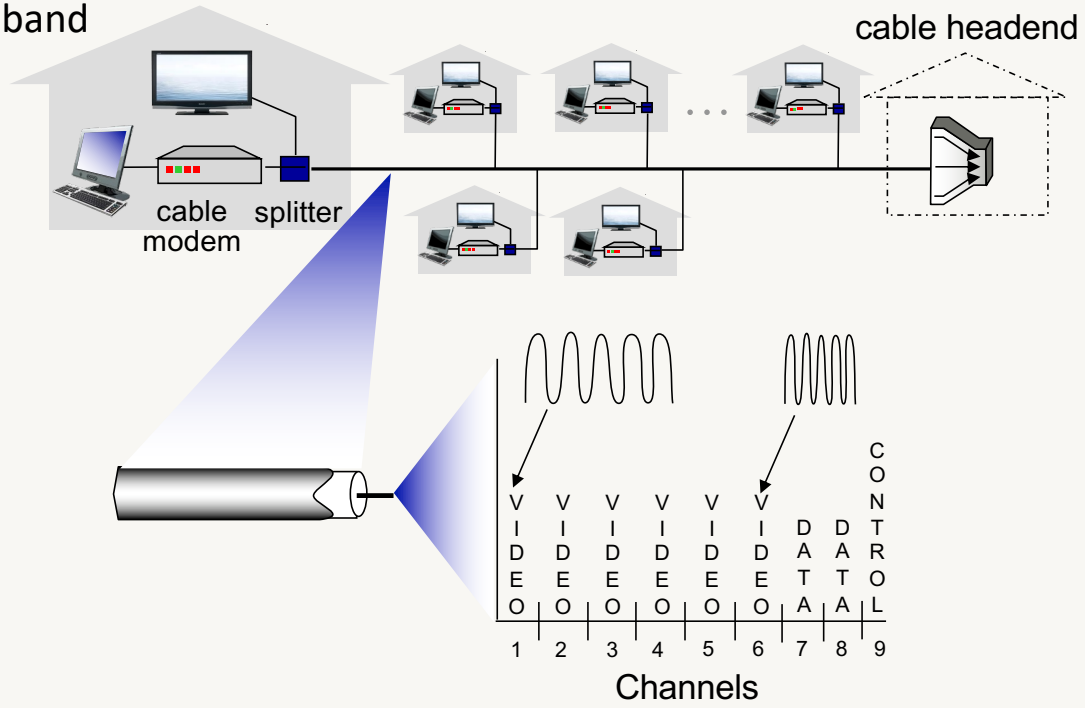
- **How to connect end systems to the edge router?**
  - Residential access networks
  - Institutional access networks (schools, companies)
  - Mobile access networks (Wi-Fi, 4G/5G)



Physical Media

Cable-Based Access

- **Frequency Division Multiplexing (FDM)**
  - Each channel is transmitted using a separate frequency band

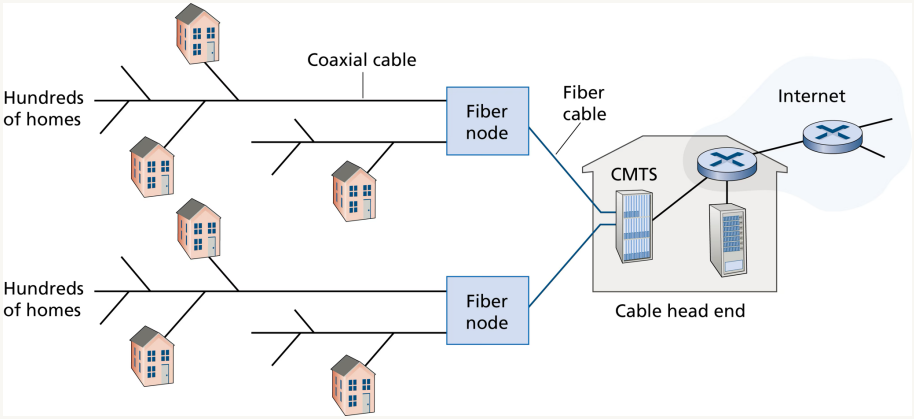


Physical Media



Cable-Based Access

- **HFC (Hybrid Fibre-Coaxial)**
  - HFC is a broadband network technology that combines optical fibre and coaxial cable to deliver Internet, TV, and phone services
  - Asymmetric transmission rates:
    - Downstream: up to 40 Mbps – 1.2 Gbps
    - Upstream: 30–100 Mbps
  - Uses a mix of cable and fibre to connect homes to the cable provider's network
  - Multiple homes share the access network to the cable headend

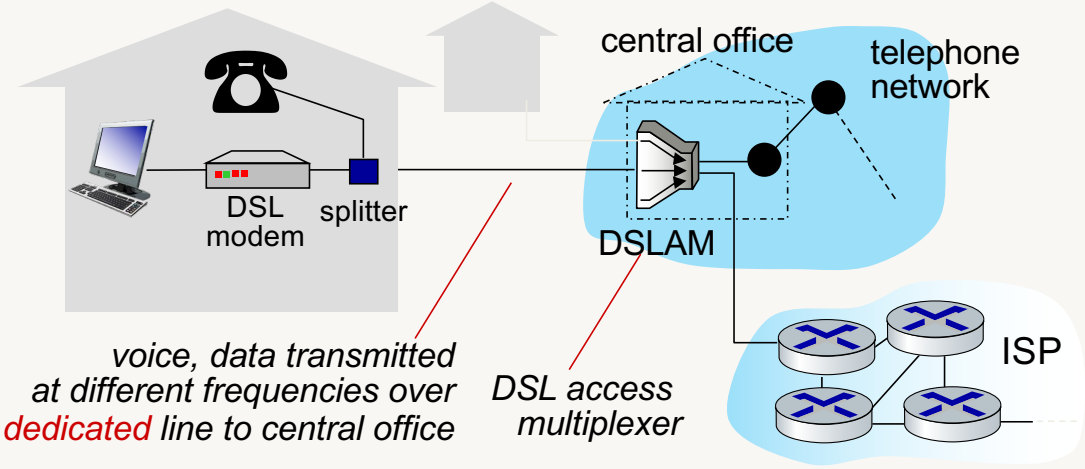


Physical Media



Digital Subscriber Line (DSL)

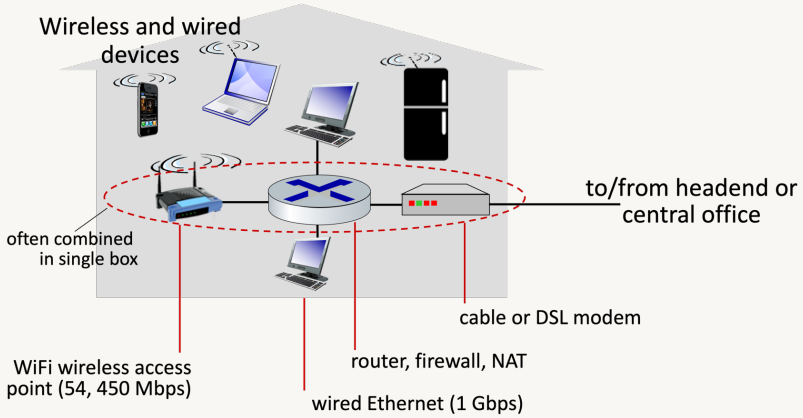
- Uses existing telephone lines to connect to the central office **DSLAM** (DSL Access Multiplexer)
  - Data over DSL line is directed to the Internet
  - Voice over DSL line is directed to the telephone network
- Provides dedicated downstream transmission rates of **24–52 Mbps**
- Provides dedicated upstream transmission rates of **3.5–16 Mbps**



Home Networks



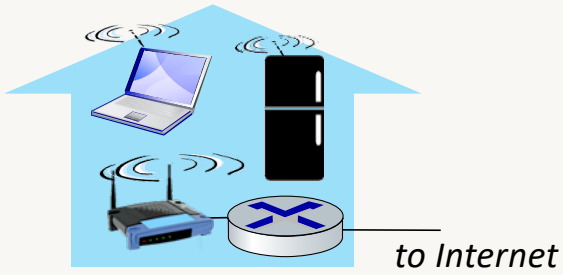
- **End devices**
  - Smartphones, computers, and smart appliances
  - Connect through Wi-Fi or Ethernet
- **Home router**
  - Provides Wi-Fi and Ethernet connectivity
  - Manages local traffic and Internet access
- **Modem**
  - Connects the home network to the ISP
  - Often combined with the router in a gateway device

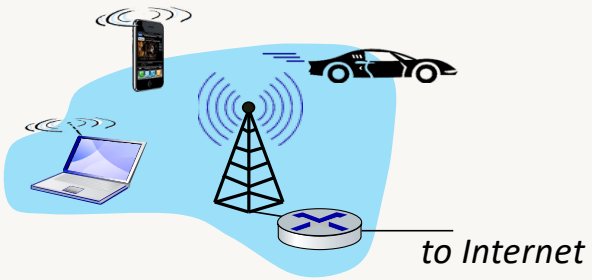


Wireless Access Networks



- **Wireless access networks** connect end systems to a router through a base station or access point
- **Wireless LANs (WLANs)**
  - Typically cover a building or an area of about 100 ft
  - Wi-Fi standards such as 802.11b/g/n offer different transmission rates
- **Cellular networks**
  - Cover larger areas through mobile providers
  - Include 4G and 5G technologies
  - Typically provide speeds ranging from tens to hundreds of Mbps

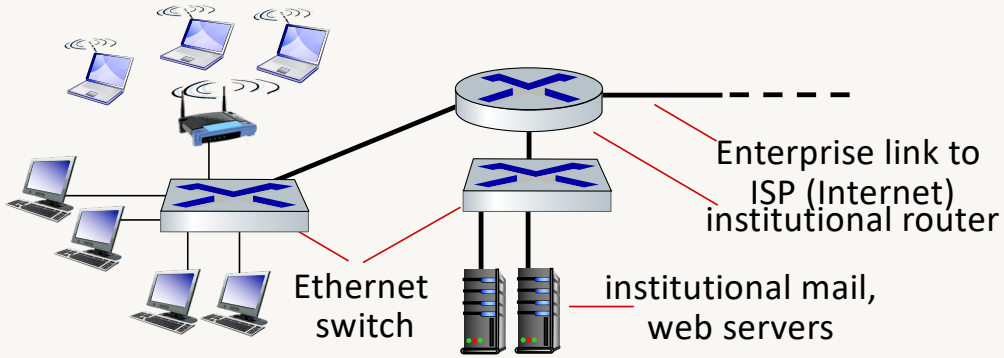




Enterprise Networks



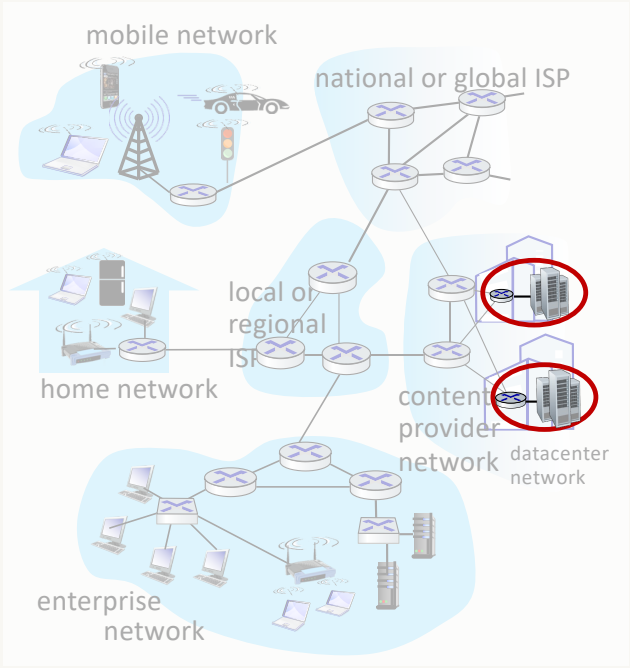
- Used by companies, universities, and similar organizations
- Combine wired and wireless link technologies with interconnected switches and routers
- Ethernet provides wired access at speeds of 100 Mbps, 1 Gbps, and 10 Gbps
- Wi-Fi access points support wireless connections at 11, 54, and 450 Mbps



Data Centre Networks



- **High-bandwidth links** (tens to hundreds of Gbps)
- Connect **hundreds to thousands of servers** within the data center
- Provide connections from servers to the **Internet**

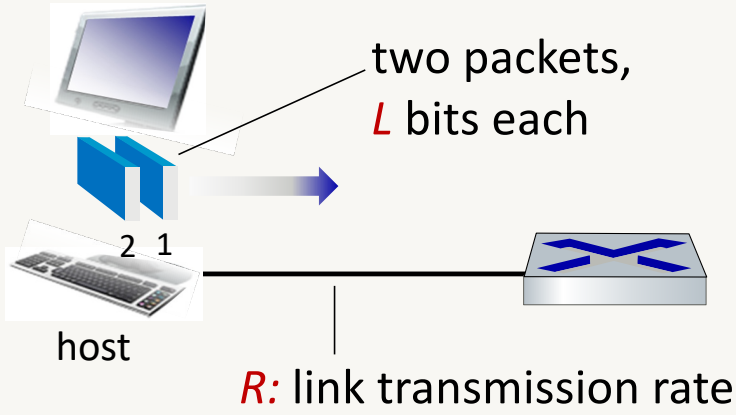


Host: Sends Packets of Data



- **Host sending function**
  - Takes an application message
  - Divides it into smaller chunks called **packets** of length *L* bits
  - Sends packets into the access network at a transmission rate *R*
  - Transmission rate is also called **link capacity** or **link bandwidth**

$$\text{packet transmission delay} = \text{time needed to transmit } L\text{-bit packet into link} = \frac{L \text{ (bits)}}{R \text{ (bits/sec)}}$$



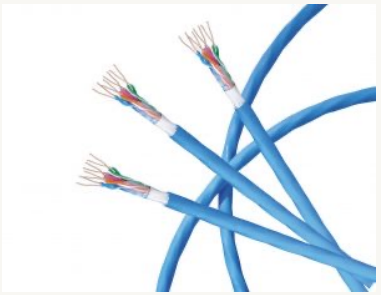
Links: Physical Media



- **Bit**: moves between a transmitter and receiver pair
- **Physical link**: the medium connecting the transmitter and receiver
- **Guided media**: signals travel through solid media such as copper, fibre, or coaxial cable
- **Unguided media**: signals travel freely through space, such as radio waves

Twisted pair (TP)

- two insulated copper wires
  - Category 5e: 100 Mbps, 1 Gbps Ethernet
  - Category 6: 10 Gbps Ethernet





Links: Physical Media



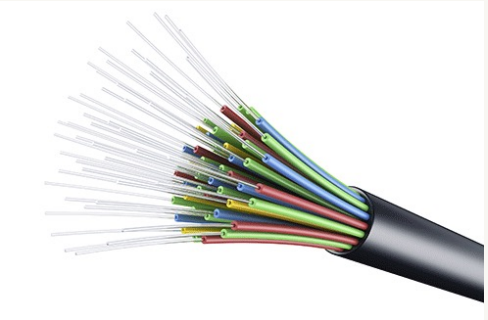
Coaxial cable:

- two concentric copper conductors
- bidirectional
- broadband:
  - multiple frequency channels on cable
  - 100's Mbps per channel



Fibre optic cable:

- glass ffibrecarrying light pulses, each pulse a bit
- high-speed operation:
  - high-speed point-to-point transmission (10's-100's Gbps)
- low error rate:
  - repeaters spaced far apart
  - immune to electromagnetic noise



Links: Physical Media



Wireless radio

- Transmits signals through electromagnetic waves without wires
- Usually uses a shared, half-duplex medium
- Affected by reflection, physical obstructions, interference, and noise

Radio Link Types

- **Wi-Fi:** High-speed access over tens of metres
- **4G/5G:** Wide-area cellular access over several kilometres
- **Bluetooth:** Short-range, low-power cable replacement
- **Terrestrial microwave:** Point-to-point wireless links
- **Satellite:** Wide-area coverage using GEO or LEO satellites
- Geostationary satellites introduce much higher delay than low-Earth-orbit systems such as Starlink.



References



- Ch. 1, Kurose and Ross
- Ch. 1, Comer



Questions?